Who drives the widening age gradient?

Heterogeneity by income, education, class, and urbanicity — and whether zero-sum beliefs match

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Summary

1. **The widening of the age gradient in redistribution preferences is concentrated among lower-income and non-college respondents.** In 1972 the young-left gradient existed only among higher-income / college respondents; the initially-flat groups converged to — and passed — them. The age×group×time triple interaction is significant for income in all three longitudinal series (GSS GOVEQINC $p < 0.001$, GSS EQWLTH $p = 0.002$, ANES VCF0809 $p < 0.001$) and for own education in the ANES ($p = 1.9 \times 10^{-8}$). Figures 1 and 2, Table 1.

2. **Today's gradient is steeper among the less-advantaged** in ANES 2024 and our Prolific survey, directionally in GSS 2021 — but **Chinoy et al. shows the opposite** (steeper for college and above-median income, precisely estimated). Unresolved; flagged below. Table 2.

3. **Zero-sum/structural beliefs do *not* mirror the subgroup pattern.** The belief gradients (esp. GSS "need wealthy family to get ahead") widened since 1987, but *in parallel* across income and parental-education groups — every belief triple interaction is insignificant (Table 3, Figure 3). For OPWLTH the null is informative: its triple's 95% CI excludes the GOVEQINC-sized preference triple. Belief widening can account for the *aggregate* steepening but not, on current evidence, for *who* steepened.

4. **Direct youth-levels test: the imprinting-on-lower-class-youth hypothesis fails — the divergence is among the old.** Tracking the *level* of under-30 support over time by SES bucket, young low- and high-SES groups rose *together* (ANES trends +0.052 vs. +0.057 SD/decade, diff. $p = 0.85$). What generates the SES-differential widening of the age gap is the *50+* bracket: older low-SES support fell or stagnated while older high-SES support rose (ANES diff. $p = 5 \times 10^{-12}$; GSS GOVEQINC $p < 0.001$). Figure 4, Table 4.

5. **Class and urbanicity add little.** Subjective class shows a stable level difference (lower/working steeper, pooled $p = 0.004$) with zero differential widening ($p = 0.97$). Urbanicity is data-limited (ANES urbanism ends 2000); GSS point estimates suggest more widening among metro residents ($p = 0.12$).

Data and specifications

Outcomes (all coded high = pro-redistribution / zero-sum view, then z-scored on the estimation sample; slopes are per 10 years of age in SD units). *Preferences:* GSS GOVEQINC ("government

should reduce income differentials," 5-pt; 9 waves 1987–2021, the paper's Figure-2 item); GSS EQWLTH (7-pt; 27 waves 1978–2024, power row); ANES VCF0809 (guaranteed jobs/income, 7-pt; 21 waves 1972–2024, the paper's Figure-2 item); Prolific `shouldconcern` (2025); Chinoy et al. `gov_outcome` (7-pt, waves pooled 2020–23 with wave FE — Chinoy waves are never used as a time dimension because of wave-6/7 sample composition changes). *Zero-sum/structural beliefs:* GSS INEQUAL3 ("inequality exists for benefit of rich," 4 waves 1987–2008 — discontinued), INEQUAL5 ("pay differences bring prosperity," flipped, 5 waves 1987–2021), CONWLTH (rich–poor conflict, 3 waves 1987–2021), OPWLTH ("need wealthy family to get ahead," 3 waves 1987–2021); Prolific `zerosumrich`; Chinoy `income_0s` (zero-sum between income classes).

Group splits (group 1 = less-advantaged throughout, so interaction signs are comparable). *Income:* GSS = family income (CONINC) below the within-year median; ANES = percentile groups (VCF0114) bottom third (0–33) vs. top third (68–100), middle third dropped; Prolific/Chinoy = household income below sample median. *Education:* own 4-yr college degree (ANES VCF0110, Prolific `edu` ≥ 6, Chinoy `education` ≥ 16); GSS extract has no own-education variable, so GSS uses *parental* college (PAEDUC ≥ 16) — a noisy proxy, which plausibly explains why GSS education interactions are attenuated relative to ANES.

Specifications. All regressions control for gender and use heteroskedasticity-robust SEs; everything is unweighted (project convention). For the *level* of the gradient (Table 2), within group g on the most recent wave (or full cross-section):

$$y_i = \alpha_g + \beta_g\,(\text{age}_i/10) + \gamma\,\text{female}_i + \varepsilon_i,$$

with the difference $\beta_1 - \beta_0$ and its p-value from the pooled age×group interaction. For *widening* (Tables 1–3), with $t10 = (\text{year} - 1990)/10$ and year FE:

$$y_{it} = \alpha_t + \beta\,\text{age10}_i + \lambda_g\,(\text{age10}_i \times t10_t) + \gamma\,\text{female}_i + \varepsilon_{it}$$

estimated within group (λ_g = change in the age slope per decade), and the **triple** = the coefficient on age10×group×$t10$ from the fully interacted pooled model. Figures show raw-scale means ±1 SE by age bracket for the oldest and newest usable wave; cells with $n < 20$ dropped.

Result 1: preference widening is a catch-up by the less-advantaged

Table 1 reports λ_g by group and the triple interaction. The income result is robust across all three longitudinal series; the education result is robust in the ANES (own education) and directional-but-insignificant in the GSS (parental proxy). Note the level result and the widening result are the same fact seen twice: pooled-over-years gradients can look *steeper for the advantaged* (they had the gradient first), which reconciles the earlier session's "class strong / income confusing" reading of pooled interactions with the endpoint figures here.

Result 2: the gradient today — three surveys vs. Chinoy

The two probability samples (ANES, GSS) and our Prolific survey agree: steeper for the less-advantaged, strongly so for ANES income. **Chinoy is the outlier in both panels**, with precisely estimated opposite-signed interactions. Candidate explanations, untested: (i) Chinoy's online panel (Lucid/Prolific-style) may select high-engagement low-income respondents differently from probability samples; (ii) its education-in-years measure and bracketed household income differ from the ANES percentile groups; (iii) genuine item difference (its 1–7 "reduce differences" stem sits between GOVEQINC and VCF0809). Worth resolving before any of Table 2 goes in the paper, since Chinoy is a headline survey.

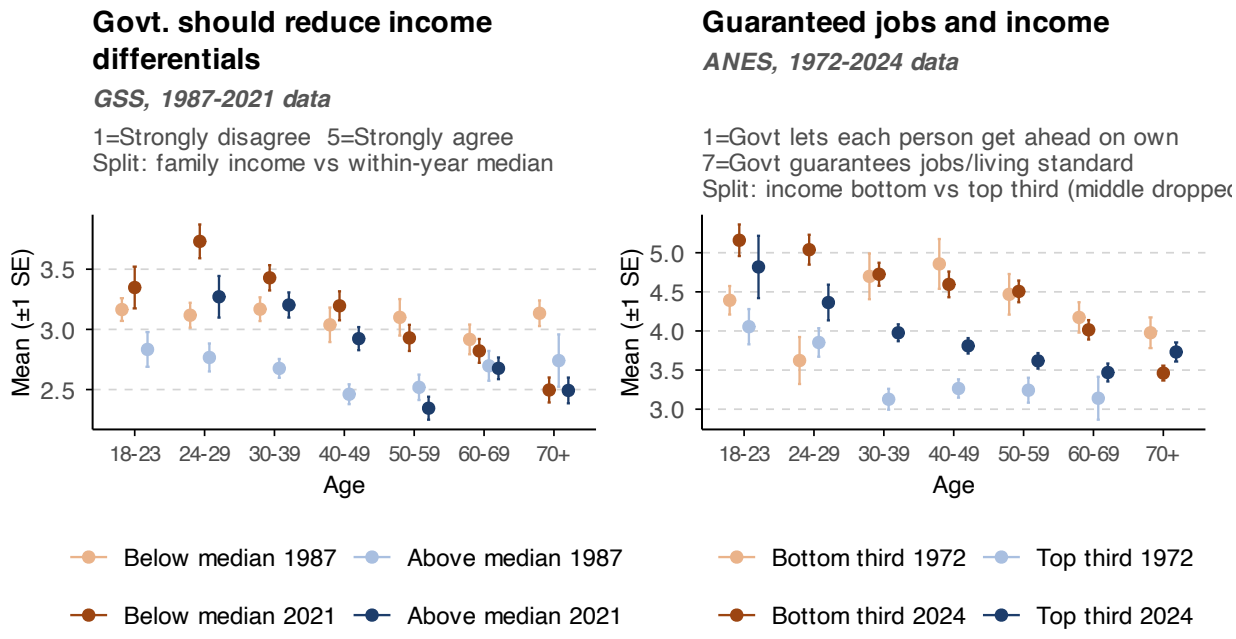


Figure 1: Redistribution support by age, income split, oldest vs. newest wave. GSS GOVEQINC (left; family income vs. within-year median) and ANES VCF0809 (right; bottom vs. top income third, middle dropped). In 1972 the age gradient exists only among the top third (light blue); by 2024 the bottom third (dark orange) is steepest.

Table 1: Widening of the age gradient by group: preferences. λ_g = change in the z-scored age slope per decade within group; triple = age×group×decade coefficient.

Series (waves)	Split	λ_{adv} (SE)	$\lambda_{\text{less-adv}}$ (SE)	Triple	p
GSS GOVEQINC (9, 1987–2021)	income	−0.010 (0.007)	−0.040 (0.006)	−0.036***	< 0.001
GSS GOVEQINC (9)	parent college	−0.022 (0.013)	−0.026 (0.006)	−0.006	0.673
GSS EQWLTH (27, 1978–2024)	income	−0.011 (0.003)	−0.024 (0.003)	−0.014***	0.002
GSS EQWLTH (27)	parent college	−0.013 (0.006)	−0.019 (0.003)	−0.008	0.212
ANES VCF0809 (21, 1972–2024)	income thirds	+0.001 (0.003)	−0.017 (0.003)	−0.020***	< 0.001
ANES VCF0809 (21)	own college	−0.006 (0.003)	−0.021 (0.002)	−0.020***	< 0.001

N: GSS GOVEQINC 11,900 (income) / 9,600 (educ.); EQWLTH 36,400 / 29,600; ANES 28,500 / 45,300. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Result 3: zero-sum beliefs widened in parallel, not differentially

Three observations. (i) The beliefs *did* widen: OPWLTH's age gradient goes from ≈ 0 in 1987 to strongly negative in 2021 in *both* income groups ($\lambda \approx -0.03$ per decade, $p < 0.01$), and INEQUAL5 shows the same in the less-advantaged groups. This is consistent with the paper's aggregate mechanism. (ii) No belief item shows *differential* widening by income — point estimates are tiny ($|\text{triple}| \leq 0.008$ for the 2021-endpoint items) with SEs ≈ 0.012, so for OPWLTH/CONWLTH the 95% CI excludes a triple as large as GOVEQINC's (−0.036) though not one as small as EQWLTH's (−0.014). The education splits are noisier (parental proxy, small college-parent cells in early waves); INEQUAL3 ends in 2008, before most of the preference widening. (iii) Levels today (Table 2B) are

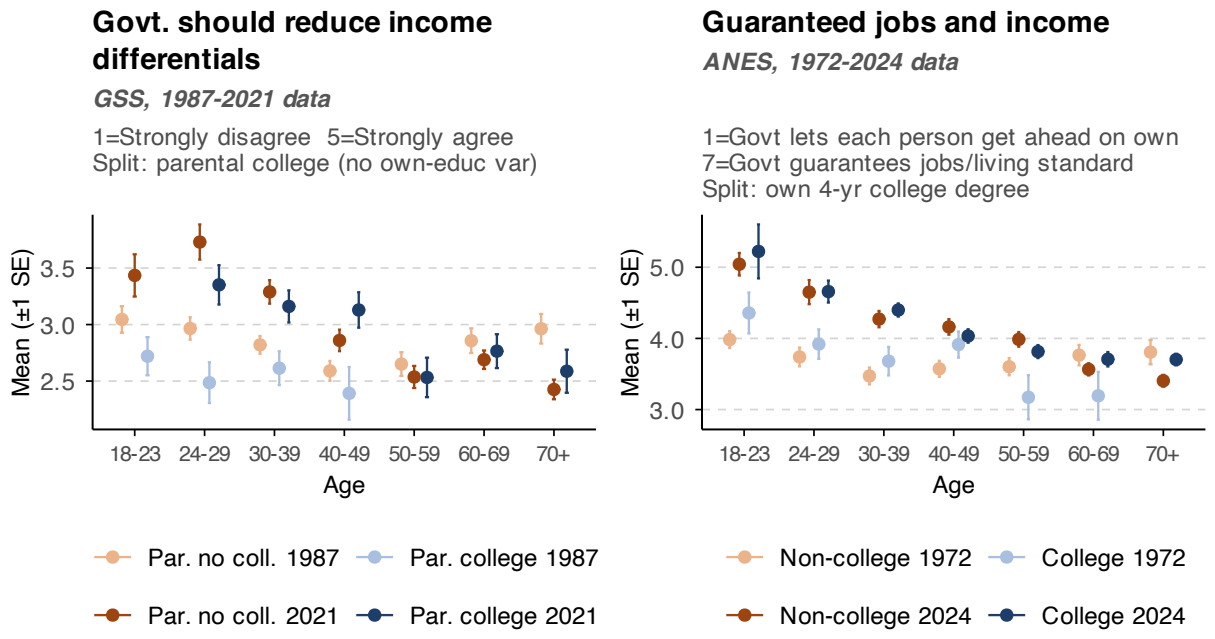


Figure 2: Same design, education split (GSS: parental college; ANES: own 4-yr degree). The 1972 gradient is a college phenomenon; by 2024 non-college respondents are at least as steep (ANES triple $p = 1.9 \times 10^{-8}$).

Table 2: The age gradient today, by group (z-scored slope per 10 years of age). Group 1 = less-advantaged. Panel B: zero-sum beliefs.

Survey, outcome	Split	β_{adv} (SE)	$\beta_{\text{less-adv}}$ (SE)	Diff.	p
Panel A: redistribution preference					
GSS GOVEQINC 2021	income	−0.130 (0.021)	−0.163 (0.019)	−0.033	0.249
GSS GOVEQINC 2021	parent college	−0.116 (0.030)	−0.156 (0.018)	−0.039	0.269
ANES VCF0809 2024	income thirds	−0.063 (0.017)	−0.158 (0.014)	−0.094***	< 0.001
ANES VCF0809 2024	own college	−0.097 (0.012)	−0.136 (0.012)	−0.039**	0.021
Prolific `shouldconcern` 2025	hh income	−0.093 (0.024)	−0.153 (0.023)	−0.060*	0.070
Prolific `shouldconcern` 2025	own college	−0.098 (0.018)	−0.147 (0.018)	−0.049*	0.053
Chinoy `gov_outcome` 2020–23	hh income	−0.142 (0.005)	−0.076 (0.005)	+0.068***	< 0.001
Chinoy `gov_outcome` 2020–23	own college	−0.134 (0.006)	−0.097 (0.005)	+0.038***	< 0.001
Panel B: zero-sum / structural beliefs					
GSS OPWLTH 2021	income	−0.098 (0.021)	−0.097 (0.020)	+0.002	0.950
GSS OPWLTH 2021	parent college	−0.137 (0.029)	−0.112 (0.019)	+0.024	0.489
Prolific `zerosumrich` 2025	hh income	−0.104 (0.023)	−0.146 (0.021)	−0.043	0.173
Prolific `zerosumrich` 2025	own college	−0.126 (0.017)	−0.124 (0.017)	+0.001	0.952
Chinoy `income_0s` 2020–23	hh income	−0.154 (0.006)	−0.089 (0.005)	+0.069***	< 0.001
Chinoy `income_0s` 2020–23	own college	−0.161 (0.007)	−0.097 (0.005)	+0.068***	< 0.001

N (recent wave / cross-section): GSS 1,330–1,570; ANES 2,840 (income) / 4,470 (educ.); Prolific 1,510–2,370; Chinoy 21,300–26,600. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

group-neutral in GSS and Prolific.

Verdict on the mechanism test: zero-sum belief widening matches the aggregate preference

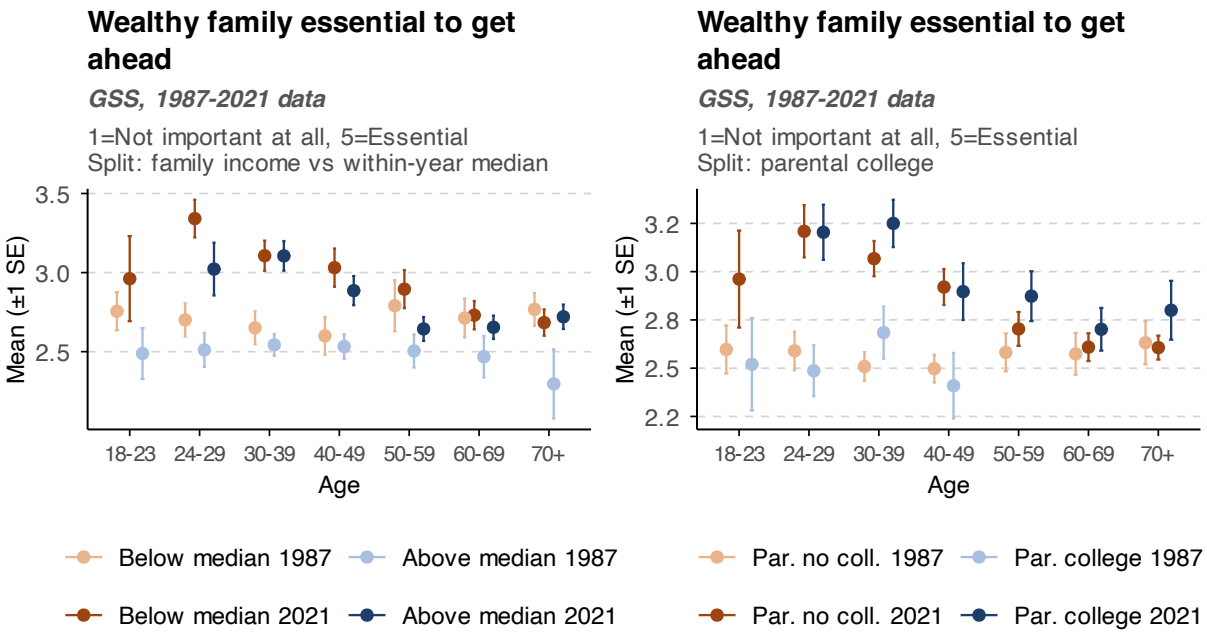


Figure 3: The structural-advantage belief (GSS OPWLTH, "how important is coming from a wealthy family to get ahead," high = essential), 1987 vs. 2021, split by income (left) and parental college (right). Both groups move from flat to steep together: the belief gradient widened, but not differentially by group.

Table 3: Widening of the age gradient by group: GSS zero-sum/structural beliefs. Same specification as Table 1; λ reported with its p-value in parentheses.

Item (waves)	Split	λ_{adv} (p)	$\lambda_{\text{less-adv}}$ (p)	Triple	p
INEQUAL3 (4, 1987–2008)	income	+0.001 (0.97)	−0.000 (0.98)	−0.002	0.940
INEQUAL3	parent college	+0.046 (0.27)	−0.008 (0.57)	−0.056	0.198
INEQUAL5 (5, 1987–2021)	income	−0.014 (0.08)	−0.019 (0.02)	−0.002	0.876
INEQUAL5	parent college	+0.016 (0.34)	−0.017 (0.02)	−0.026	0.140
CONWLTH (3, 1987–2021)	income	−0.008 (0.39)	+0.000 (0.96)	+0.008	0.564
CONWLTH	parent college	−0.010 (0.62)	+0.002 (0.83)	+0.010	0.634
OPWLTH (3, 1987–2021)	income	−0.028 (0.003)	−0.032 (< 0.001)	−0.003	0.794
OPWLTH	parent college	−0.027 (0.14)	−0.040 (< 0.001)	−0.016	0.413

High = zero-sum/conflict/structural view throughout (INEQUAL5 flipped: high = rejects "pay differences bring prosperity"). N per row 2,000–6,000 (see `_build/zerosum_by_group.csv`). ANES has no zero-sum item.

widening but not its subgroup composition. If zero-sum beliefs drive the widening, the group-specific catch-up must run through something else — e.g. a rising *pass-through* of the same beliefs into preferences among the less-advantaged, or a belief margin GSS does not measure. A direct test: interact the belief with group in a preference regression by era (does the OPWLTH→GOVEQINC slope grow more for low-income respondents?), or field the subgroup question in Prolific wave 2.

Result 4: young SES groups rose together — the divergence is among the old

A more direct test of the imprinting hypothesis (proposed by Niclas): if the widening reflects imprinting on lower-class youth, then *among people in the impressionable age range*, low-SES respondents should show a stronger rise in support than high-SES respondents. Table 4 tracks the level of support within the paper's under-30 bracket (18–29), and within 50+, as a linear trend per decade (z-scored outcome, pooled scaling so trends are comparable; gender control, robust SEs).

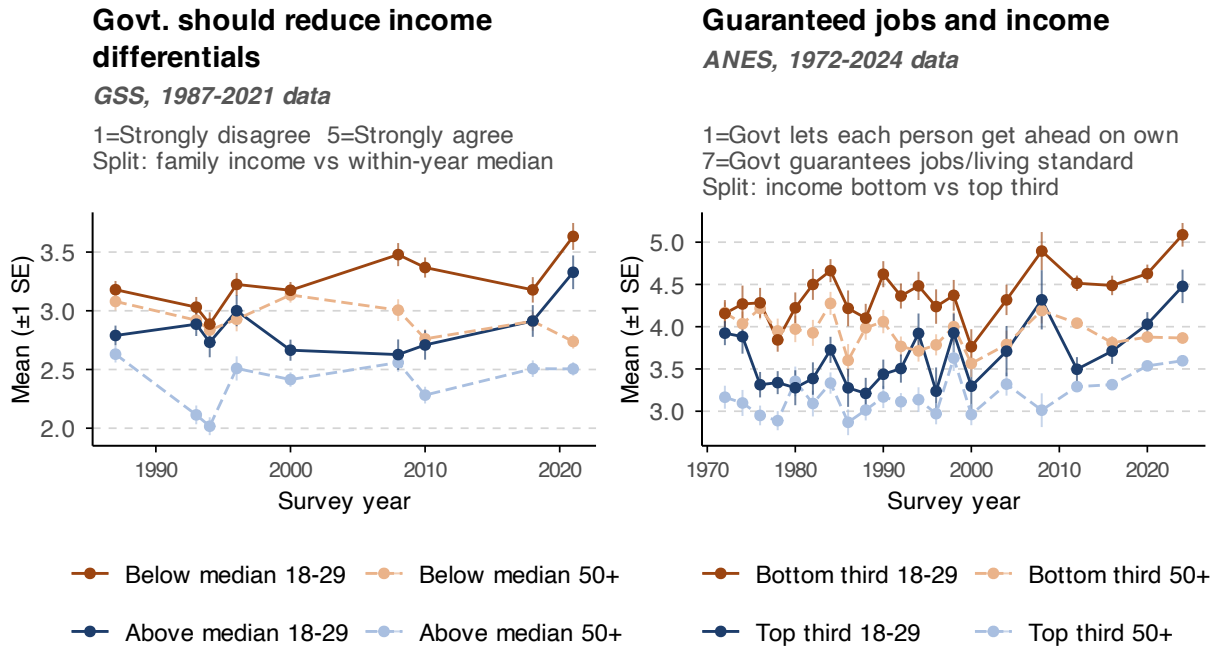


Figure 4: Support for redistribution by survey year: age 18–29 (solid, dark) vs. 50+ (dashed, light), split by income. The two young series rise roughly in parallel; the divergence is between the two *old* series — older below-median-income support drifts down while older above-median support rises.

Three conclusions. **(i) The lower-class-youth imprinting hypothesis is not supported.** Young low- and high-income groups rose in near-lockstep (ANES: +0.052 vs. +0.057; GSS: directionally faster for the young poor but $p \geq 0.13$), and on the parental-education split the *parent-college* young rose *fastest* (EQWLTH diff $p = 2.5 \times 10^{-5}$ in the opposite direction). **(ii) The SES-differential widening of the age gap documented in Table 1 lives in the 50+ bracket:** older low-SES Americans turned *against* redistribution (in relative and often absolute terms) while older high-SES Americans warmed to it — e.g. ANES $\lambda_{\text{adv}} \approx 0$ arises because young and old advantaged respondents rose at the same rate, while $\lambda_{\text{less-adv}} < 0$ arises because old less-advantaged respondents fell as their young counterparts rose. This looks less like youth imprinting and more like the well-documented realignment of older working-class voters, and it reframes what needs explaining. **(iii) The zero-sum belief again moves in parallel:** OPWLTH rose among all young groups regardless of SES — consistent with beliefs driving the *common* youth shift (the paper's mechanism), with the old-side realignment as a separate phenomenon. *Caveat:* education-group composition drifts over five decades (the no-college young grow increasingly selected), so the ed-

Table 4: Trend in the level of support, per decade (z units), by age bracket and group. Diff = less-advantaged minus advantaged trend.

		Age 18–29			Age 50+		
Series	Split	adv.	less-adv.	diff (p)	adv.	less-adv.	diff (p)
GSS GOVEQINC	income	+0.054	+0.103	+0.048 (0.20)	+0.035	−0.058	-0.093^{***}
GSS GOVEQINC	parent college	+0.132	+0.091	−0.041 (0.36)	+0.074	−0.022	-0.097^{**}
GSS EQWLTH	income	+0.058	+0.082	+0.025 (0.13)	+0.031	−0.009	-0.040^{***}
GSS EQWLTH	parent college	+0.138	+0.055	-0.084^{***}	+0.118	−0.000	-0.119^{***}
ANES VCF0809	income thirds	+0.052	+0.057	+0.003 (0.85)	+0.054	−0.017	-0.070^{***}
ANES VCF0809	any college	+0.085	+0.090	+0.004 (0.79)	+0.060	−0.002	-0.061^{***}
GSS OPWLTH (belief)	income	+0.145	+0.128	−0.018 (0.75)	+0.047	+0.001	−0.046 (0.16)

ANES education among 18–29 uses *any college exposure* (VCF0110 ≥ 3; a 4-yr degree is censored at these ages); the degree-based split at 25–34 gives the same null (diff −0.012, $p = 0.37$). SEs 0.006–0.05, in `_build/youth_levels.csv`. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

ucation rows understate/overstate selection effects; the income split is a within-year median and immune to share drift — lean on the income rows. Household income for 18–29-year-olds partly reflects parental/coresident households.

Confirmation across other redistribution outcomes

Table 5 repeats the young/old trend-by-group comparison for every other redistribution-like longitudinal outcome available (ANES services/spending and government health insurance; GSS HELPPOOR; WVS US; ISSP US Social Inequality and Role of Government modules). The pattern replicates: **the old-side difference is negative in 12 of 14 cells and significant in 8** (the two exceptions are the data-poor WVS income split, which ends in 2006), **while no young-side income difference is distinguishable from zero anywhere**. Where young-side education differences are significant they favor the *advantaged* young rising faster (as with EQWLTH above; same composition caveat). Whether old-side convergence comes from the old poor cooling absolutely (health insurance, ISSP jobs) or the old rich warming faster (services, ISSP v27) varies by item; the *relative* convergence is uniform.

Class and urbanicity (negative results, for the record)

Subjective class (GSS CLASS_ lower/working vs. middle/upper; ANES VCF0148A, discontinued 2016): a stable level difference — lower/working steeper by ≈ 0.014 SD/decade-of-age pooled ($p = 0.004$, GSS EQWLTH) — but no differential widening whatsoever (triple $p = 0.97$ GSS; $p = 0.19$ ANES through 2016). Class tells us who has the gradient, not who changed. *Urbanicity*: ANES's urbanism variable ends in 2000 (before the action); GSS current residence (metro vs. non-SMSA) shows more widening among metro residents (endpoint triple $p = 0.12$, rural 2021 $n = 212$), while childhood urbanicity (RES16, EQWLTH) shows *rural-raised* respondents catching up ($p = 0.017$) — i.e. the same convergence signature, but current-vs-childhood measures disagree on direction and neither is strong enough to feature.

Table 5: Robustness across outcomes: difference in trend (less-advantaged minus advantaged, z units per decade) within each age bracket.

Series (span)	Split	Young 18–29	Old 50+
ANES VCF0839 services (1980–2020)	income thirds	+0.018 (0.44)	−0.030** (0.03)
	any college	−0.045** (0.02)	−0.062***
ANES VCF0806 govt health ins. (1970–2020)	income thirds	−0.027 (0.16)	−0.095***
	any college	−0.010 (0.51)	−0.069***
GSS HELPPOOR (1975–2024)	income	+0.024 (0.17)	−0.025** (0.04)
	parent college	−0.046** (0.03)	−0.112***
WVS US E037 govt resp. (1995–2017)	income (≤2006)	+0.083 (0.63)	+0.114 (0.27)
	educ (upper)	−0.047 (0.40)	−0.078* (0.06)
WVS US E035 equality (1995–2017)	income (≤2006)	−0.071 (0.64)	−0.078 (0.47)
	educ (upper)	−0.041 (0.48)	−0.057 (0.21)
ISSP V27 reduce diff. (1987–2019)	income terciles	−0.093 (0.13)	−0.123***
	university	−0.112** (0.04)	−0.223***
ISSP V50 jobs (1985–2016)	university	+0.068 (0.34)	−0.030 (0.47)
ISSP V56 reduce diff. (1985–2016)	university	−0.058 (0.37)	−0.168***

p-values in parentheses (< 0.01 omitted). ISSP income = the derived household-income grouping INCOME, "Low" vs. "High" with "Middle" dropped (these are *not* terciles — the US cells are uneven, e.g. 208/144/117 in 1987); WVS income (X047CS) missing after 2006. Full trends by group in `_build/oldside_confirm.csv` (script `29_oldside_confirm.R`). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Pointers

Scripts (all in `cleaned/analysis/`, exploratory, not in `run_all.R`): figures `24_group_widening_pointrange.R`, `27_zerosum_figure.R`; estimates `25_zerosum_by_group.R`, `26_heterogeneity_tables.R` (this memo's tables), `28_youth_levels.R` (Result 4), plus the earlier heterogeneity sweep `18/19/21_*.R`. CSVs in `cleaned/output/_build/`: `het_table_levels.csv`, `het_table_widening.csv`, `zerosum_by_group.csv`, `youth_levels.csv`, `oldside_confirm.csv`. Additional figures (class, urban/rural) in `_build/` (`group_widening_class`, `_urban`). All estimates unweighted; computed September 21, 2026.